Magnetic & Electric Fields Mini-Assessment

TOTAL: 49

_____

Atoms consist of two types of charged particles.

1. **State** the name of the positively charged particle.

_____

State the name of the negatively charged particle.

_____

State the name of the particle with no charge.

_____

(3 marks)

2. **List** three places you could use a magnet.

a. _____
_____

b. _____
_____

c. _____
_____

(3 marks)

3. **State** which term correctly completes each sentence.

a. A magnetic force is a **contact/non contact** force.

b. As you get closer to a magnet, the size of the magnetic force **increases/decreases.**

c. A north pole of one magnet is attracted to the **south/north** pole of another magnet.

d. A magnet is **strongest/weakest** at its poles.

(4 marks)

4. If a plastic ruler loses electrons when it is rubbed on a piece of woollen fabric, **state** whether it has become positively or negatively charged.

_____
_____

(1 mark)

5. **Name** the three metals that are attracted to a magnet.

_____
_____

(3 marks)

6. **Calculate** the average number of paperclips picked up by an electromagnet with different number of turns of wire around them and different voltage running through them.

25 Turn Electromagnet

Battery Voltage	Number of Paperclips			Average
	Trial 1	Trial 2	Trial 3	
1.5	5	7	7	
3.0	12	10	13	
4.5	14	17	15	
6.0	20	26	25	

(4 marks)

50 Turn Electromagnet

Battery Voltage	Number of Paperclips			Average
	Trial 1	Trial 2	Trial 3	
1.5	10	14	17	
3.0	28	24	26	
4.5	30	34	33	
6.0	44	50	49	

(4 marks)

7. **Graph** the average number of paperclips for the 25 and 50 turn electromagnet on the same graph paper as a line graph.

(7 marks)

a) Which voltage can pick up the most paperclips?

_____

(1 mark)

b) Which electromagnet can pick up the most paperclips?

_____

(1 mark)

c) What is the highest number of paperclips picked up in any trial?

_____

(1 mark)

d) What is the range in the number of paperclips picked up by either magnet.

_____

(1 mark)

8. **Identify** each of these as attracting or repelling.

a. North and North. _____

b. North and South. _____

c. South and South. _____

(3 marks)

9. **Classify** the following materials as conductors or insulators.

a) aluminium foil _____

b) foam cup _____

c) plastic ruler _____

d) copper wire _____

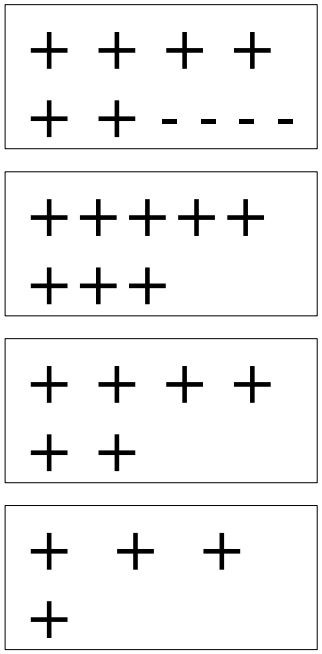
e) silver bracelet _____

f) rubber hose _____

(6 marks)

10. **Identify** the following as positive, negative or neutral.

a)



(4 marks)

11. If you rub a CD with a cloth to clean it, you may notice dust drifting towards it and landing on its surface. **Explain** why this is happening.

_____
_____

_____
_____

_____
_____

_____
_____

(2 marks)

12. If a comb pulled through your hair becomes negatively charged, then predict the charge of your hair.

_____
_____

(1 marks)